

CYGNUS ATRATUS

In his *Treatise on Human Nature*, the Scots philosopher David Hume posed the issue in the following way (as rephrased in the now famous *black swan problem* by John Stuart Mill): *No amount of observations of white swans can allow the inference that all swans are white, but the observation of a single black swan is sufficient to refute that conclusion.*

Hume had been irked by the fact that science in his day (the eighteenth century) had experienced a swing from scholasticism, entirely based on deductive reasoning (no emphasis on the observation of the real world) to, owing to Francis Bacon, an overreaction into naïve and unstructured empiricism. Bacon had argued against “spinning the cobweb of learning” with little practical result. Science had shifted, thanks to Bacon, into an emphasis on empirical observation. The problem is that, without a proper method, empirical observations can lead you astray. Hume came to warn us against such knowledge, and to stress the need for some rigor in the gathering and interpretation of knowledge – what is called epistemology (from *episteme*, Greek for learning). Hume is the first modern *epistemologist*, (epistemologists are often called methodologists or philosophers of science). What I am writing here is not strictly true, for Hume said things far worse than that; he was an obsessive skeptic and never believed that a link between two items could be truly established as being causal. But we will tone him down a bit for this book.

NIEDERHOFFER, VICTORIAN GENTLEMAN

It is worth noting that finance has its Francis Bacon in the person of Victor Niederhoffer. He was the very first to stand against the *cobweb of learning* of the University of Chicago and the efficient market religion of the 1960s, when it was at its worst. In contrast with the scholasticism of financial theorists, he looked at data in search of anomalies – and found enough of them to be able to conduct a successful career in randomness and deliver an insightful book, *The Education of a Speculator*. Since then, an entire industry of such operators, called “statistical arbitrageurs”, flourished, the major and most successful ones

were initially his trainees. While Niederhoffer had a publicized hiccup, some of his trainees fared well because they added rigor and *methodology* to their statistical inference. In other words, Niederhoffer’s empiricism missed just a modicum of methodology.

I have to admit that for all my intellectual disagreements with him I have been inspired by his empiricism and owe him a large share of my intellectual growth. I experienced a jump in my trading style in 1996, when Victor blurted to me that any “testable” statement should be *tested* (it was so obvious but I had not done it until then). His advice went straight home. A testable statement is one that can be broken down into quantitative components and subjected to statistical examination. For instance, a conventional-wisdom style statement like

accidents happen closer to home

can be tested by taking the average distance between the accident and the domicile of the driver (if, say, about 20% of accidents happen within a 12-mile radius). However, one needs to be careful in the interpretation; a naïve interpreter of the result would tell you that you are more likely to have an accident if you drive in your neighborhood than if you did so in remote places, which is an example of naïve empiricism. Why? Because accidents may happen closer to home simply because people spend their time driving close to home (if people spend 20% of their time driving in a 12-mile radius).

Since that very same day I have not made a single testable proposition without testing it, thanks to the computer which I rarely use for non-computational tasks. However, the differences between Victor Niederhoffer and myself remain immense; I can use data to disprove a proposition, never to prove one. I can use history to refute a conjecture, never to affirm it. For instance, the statement

The market never goes down 20% in a given 3-month period

can be tested, but is completely meaningless if verified. I can quantitatively reject the proposition by finding counterexamples, but it is impossible for me to accept it simply because in past data the market never went down 20% in any 3-month period.